5	**(a)** E = stress / strain	B1	[1]
	(b) (i) 1. diameter / cross sectional area / radius 2. original length	B1	[1]
	(ii) measure original length with a <u>metre</u> ruler / tape	B1	
	measure the <u>diameter</u> with micrometer (screw gauge) *allow digital vernier calipers*	B1	[2]
	(iii) energy = ½ *Fe* or area under graph or ½ kx^2	C1	
	$= ½ \times 0.25 \times 10^{-3} \times 3 = 3.8 \times 10^{-4}$ J	A1	[2]
	(c) straight line through origin below original line	M1	
	line through (0.25, 1.5)	A1	[2]
6	**(a)** two waves travelling (along the same line) in opposite directions overlap/meet	M1	